1899. Merge Triplets to Form Target Triplet

Medium · Greedy

leetcode.com/problems/merge-triplets-to-form-target-triplet

Problem

- Merging two triplets takes the element-wise max. Can some sequence of merges produce exactly target?

Intuition

- A triplet with any value greater than target in some position would overshoot forever, so ignore it.
- Among the safe triplets, each target coordinate just needs to be matched exactly by at least one of them.

Brute Force

- Try all subsets of triplets: O(2^n).

Optimal Approach

- Found[3] = false.
- For each triplet t with t[k] <= target[k] for all k: for each k, if t[k] == target[k], found[k] = true.
- Return true only if all three are found.

Worked Example

```
triplets [[2,5,3],[1,8,4],[1,7,5]], target [2,7,5].

[2,5,3] safe -> matches x (2)
[1,8,4] 8 > 7 -> skip
[1,7,5] safe -> matches y (7) and z (5)
all matched -> true
```

Java Solution

```java
public class MergeTripletsToFormTarget {
    public boolean mergeTriplets(int[][] triplets, int[] target) {
        boolean[] found = new boolean[3];
        for (int[] t : triplets) {
            if (t[0] > target[0] || t[1] > target[1] || t[2] > target[2]) continue;
            for (int k = 0; k < 3; k++) if (t[k] == target[k]) found[k] = true;
        }
        return found[0] && found[1] && found[2];
    }
}
```

Complexity

- Time: O(n). Space: O(1).

Pitfalls

- A single safe triplet can satisfy several coordinates.